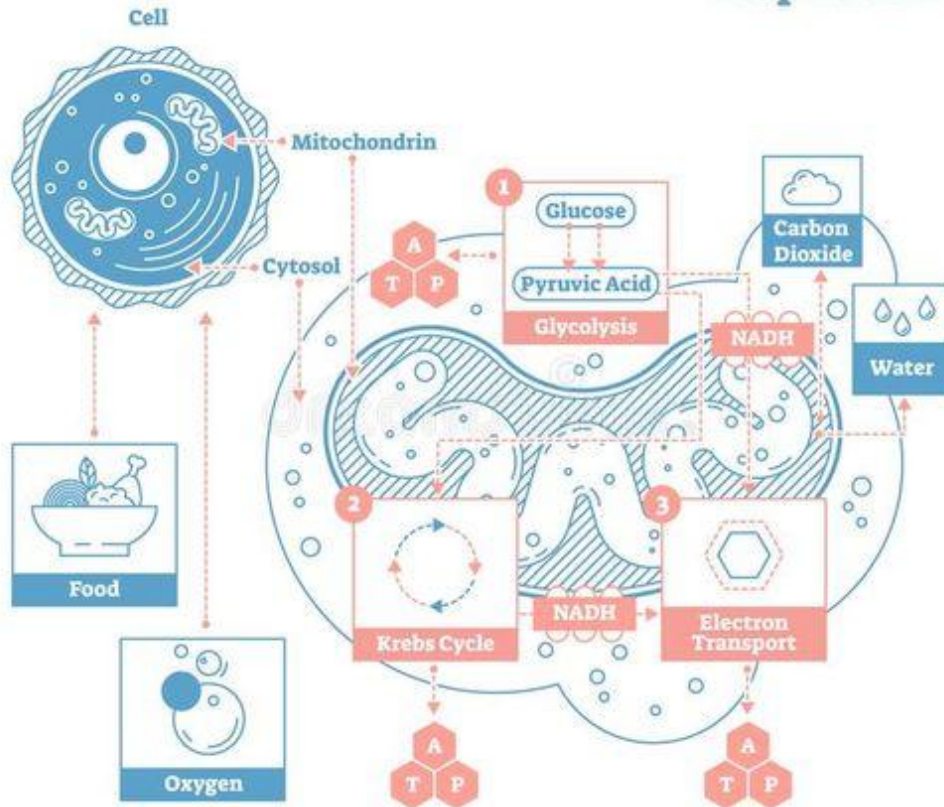


Cellular Respiration



Chapter 7 Energy & Metabolism

Course: BIO 101,
Introduction to Biology

What is Energy

- Energy is the 'capacity to do work'.
- Anything our body does, it needs a certain level of energy to function.
Examples of activities that require energy include:

- ✓ walking
- ✓ running
- ✓ exercising
- ✓ doing homework
- ✓ eating
- ✓ sleeping
- ✓ breathing



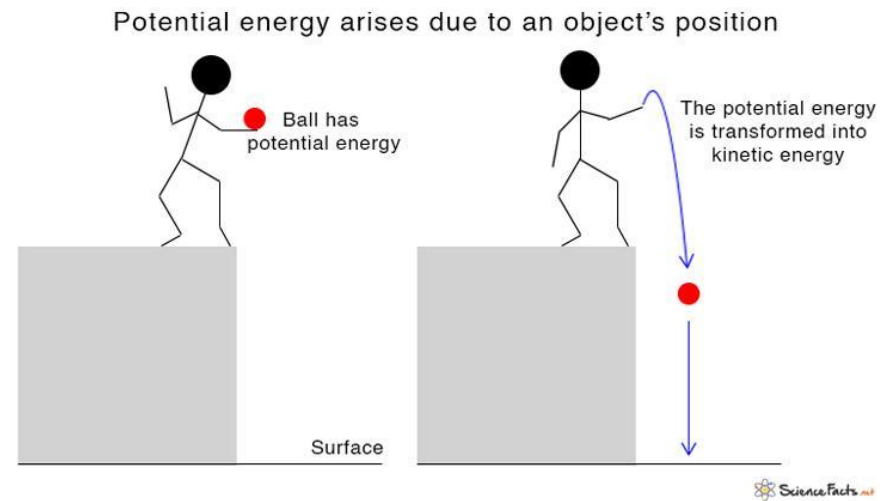
Energy exists in different forms!

- **Potential energy:** energy that is stored in an object at rest, based on the position it is in.

- E.g. a ball raised above ground

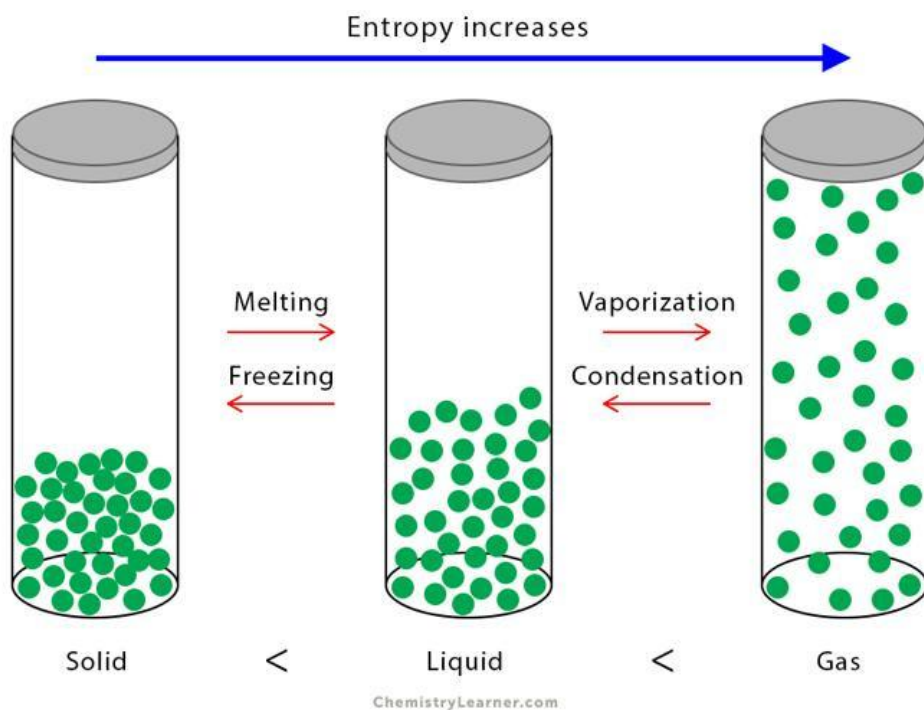
- **Kinetic energy:** Energy an object has because of its motion.

- E.g. a ball that has been thrown



Thermodynamics

- Heat is a form of energy, called **thermal energy**.
- The study of the conversion of thermal energy to and from other types of energy is called thermodynamics.
- All metabolic reactions release some amount of heat.



- **First Law of Thermodynamics**

Energy cannot be created or destroyed, can be converted from one form to another.

- **Second Law of Thermodynamics**

Energy tends to spread out or disperse spontaneously from high energy forms to forms lower in energy. The tendency of entropy to increase is the second law of thermodynamics.

- **Entropy**

Measure of how much the energy of a particular system has become dispersed/ number of possible states, a system can have.

Metabolism

Metabolism is collectively all the chemical reactions that take place within each cell of a living organism for,

- providing energy to vital processes
- for synthesizing new organic material.

- Cells store energy by building organic molecules (within bonds) and retrieve energy by breaking them down.

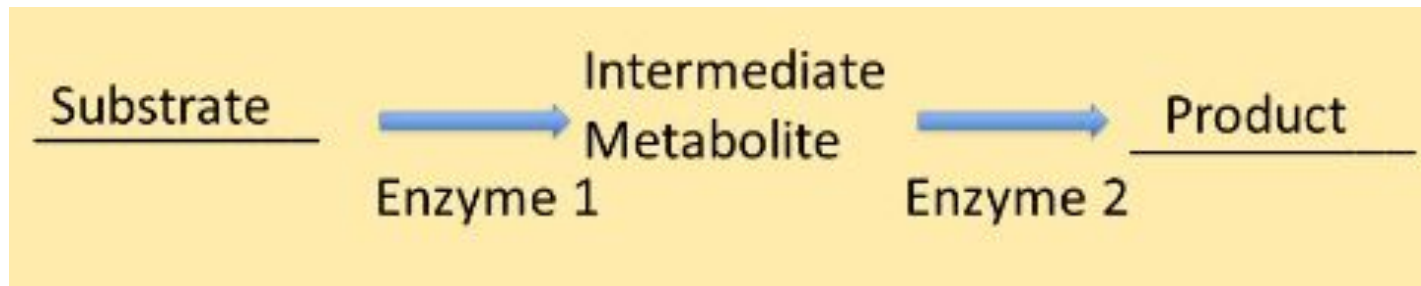
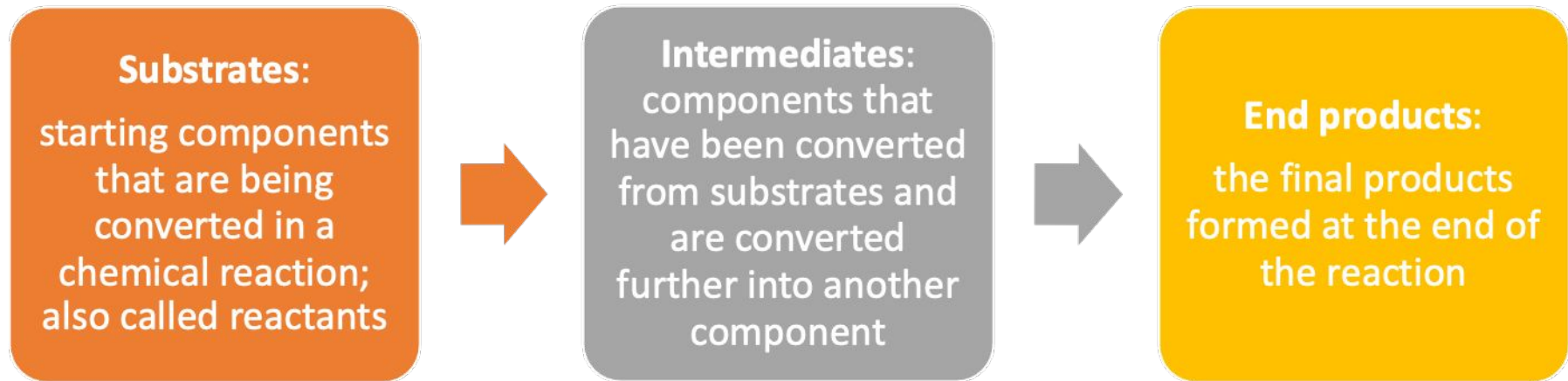


Fig: A simple Metabolic pathway

Participants in metabolic pathways

A reaction occurs when components undergo the following changes:



Several biological molecules assist in completion of a reaction, which are:

- **Energy Carriers:** ATP
- **Enzymes:** Biological molecules that speed up a reaction
- **Cofactors:** Non-protein chemical compounds that aid in an enzyme's activity



Types of Metabolic pathways

- Anabolic Pathway
- Catabolic Pathway

Metabolic pathways

Anabolic: Small molecules are assembled into large ones. *Energy is required.*



Catabolic: Large molecules are broken down into small ones. *Energy is released.*



Energy in the Molecules of Life

Chemical Energy: Every chemical bond holds a certain amount of energy.

- The particular amount of energy held by a bond depends on which elements are taking part in it. For example, two covalent bonds—one between an oxygen and a hydrogen atom in a water molecule, the other between two oxygen atoms in molecular oxygen (O_2)—both hold energy, but different amounts.

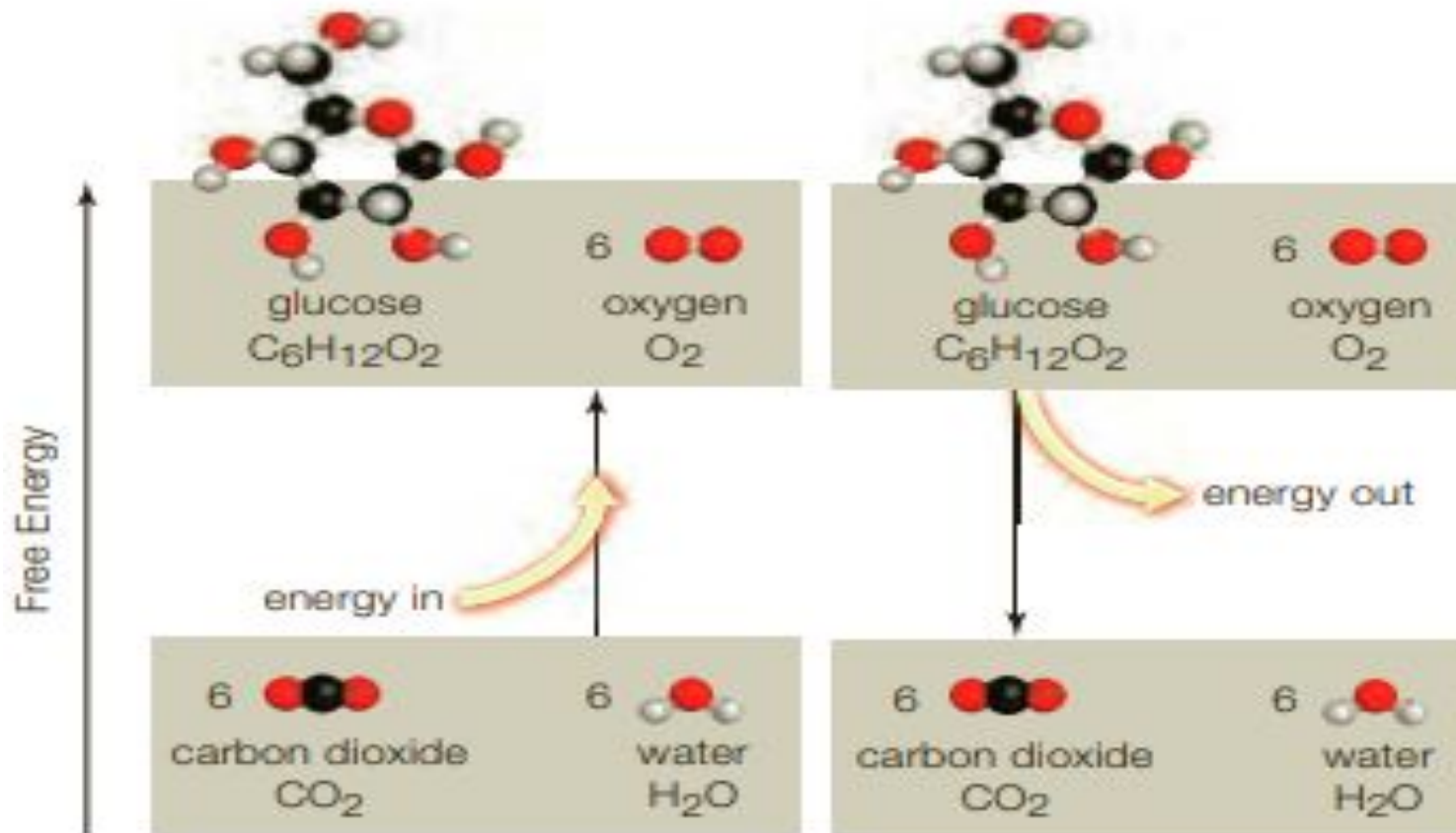
- Energy changes in living body/cells occur mostly via two types reactions:

1. Endergonic Reactions

- Energy input required.
- Product has more energy than starting substances.
- Anabolic reactions are endergonic.
- Energy “in”

2. Exergonic Reactions

- Energy is released.
- Products have less energy than starting substance.
- Catabolic reactions are exergonic.
- Energy “out”



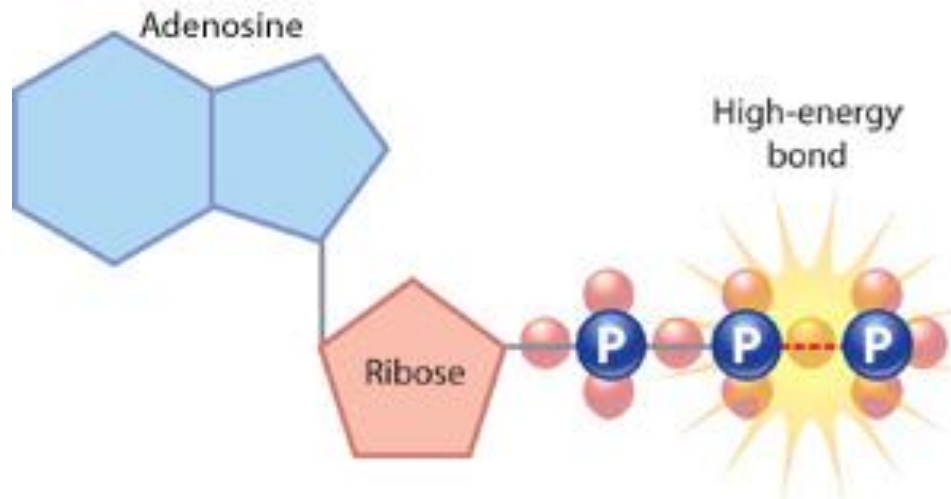
A Endergonic reactions convert molecules with lower free energy to molecules with higher free energy, so they require a net energy input.

B Exergonic reactions convert molecules with higher free energy to molecules with lower free energy, so they end with an energy release.

Figure 5.6 The ins and outs of energy in chemical reactions.

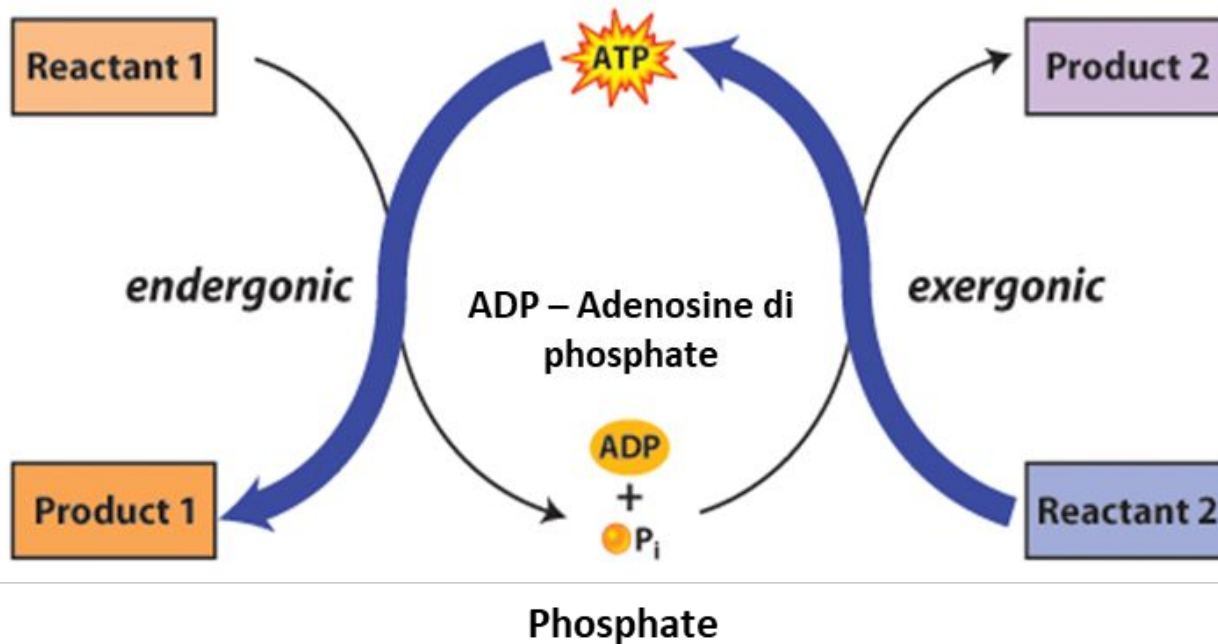
What is ATP?

- Adenosine triphosphate
- ATP is the energy-carrying molecule found in cells of living organisms.
- It is composed of :
 - One adenosine (adenine + ribose sugar) & three phosphate groups
 - One adenosine + two phosphate=ADP (Adenosine diphosphate)



The Role of ATP?

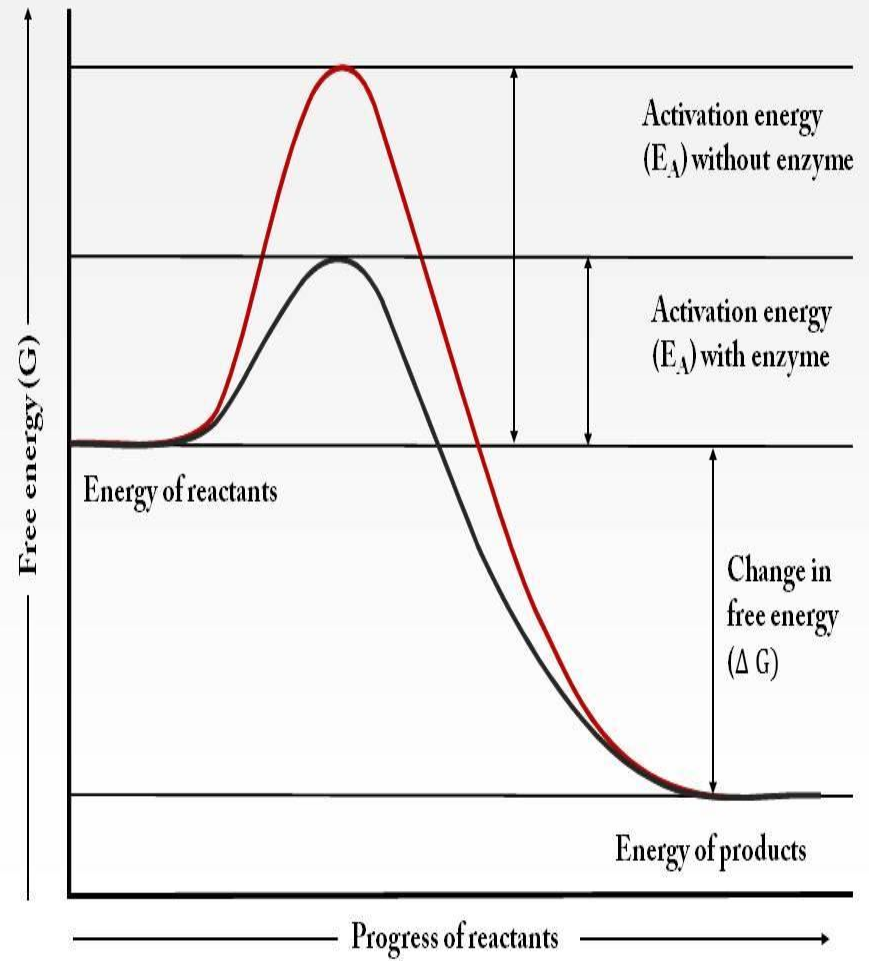
- Cells “earn” energy (ATP) in exergonic reactions
- Cells “spend” energy (ATP) in endergonic reactions



Activation Energy

- For a reaction to occur, an energy barrier must be overcome.
- Activation energy is the minimum amount of energy needed for a chemical reaction to occur.
- Its like a hill that reactants must climb before they can coast down the other side to become products
- Enzymes make the energy barrier (activation energy) smaller, hence the reaction runs faster.
- Both endergonic and exergonic reactions have activation energy, but the amount varies with the reaction.

Enzymes & Activation Energy



Membrane Transport Mechanism

- Nutrients and other compounds are required for energy production. So, communication between the cell and the extracellular medium is necessary.
- For the exchange of substances in and out of cells, special transport systems are required.
- This is made possible by the semipermeable phospholipid bilayer, the main component of cell membrane. The phospholipid bilayer regulates the entry and exit of specific molecules through different mechanisms.

- **Types of transport system**

- ☐ Diffusion across lipid bilayer
- ☐ Passive transport (transport protein)
- ☐ Active transport (transport protein)
- ☐ Endocytosis
- ☐ Exocytosis

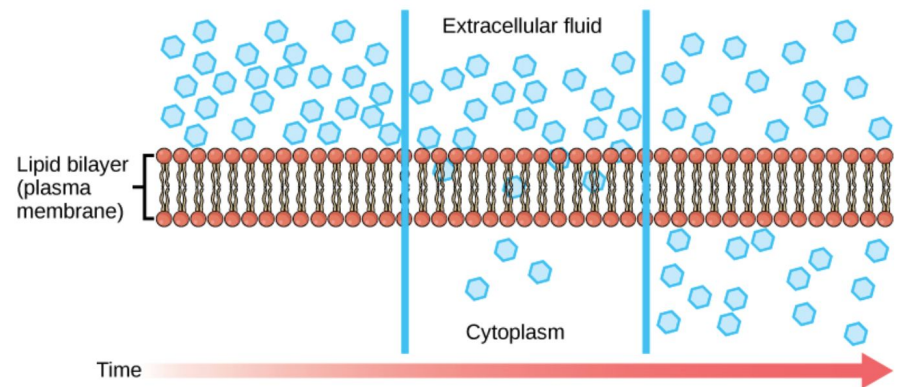
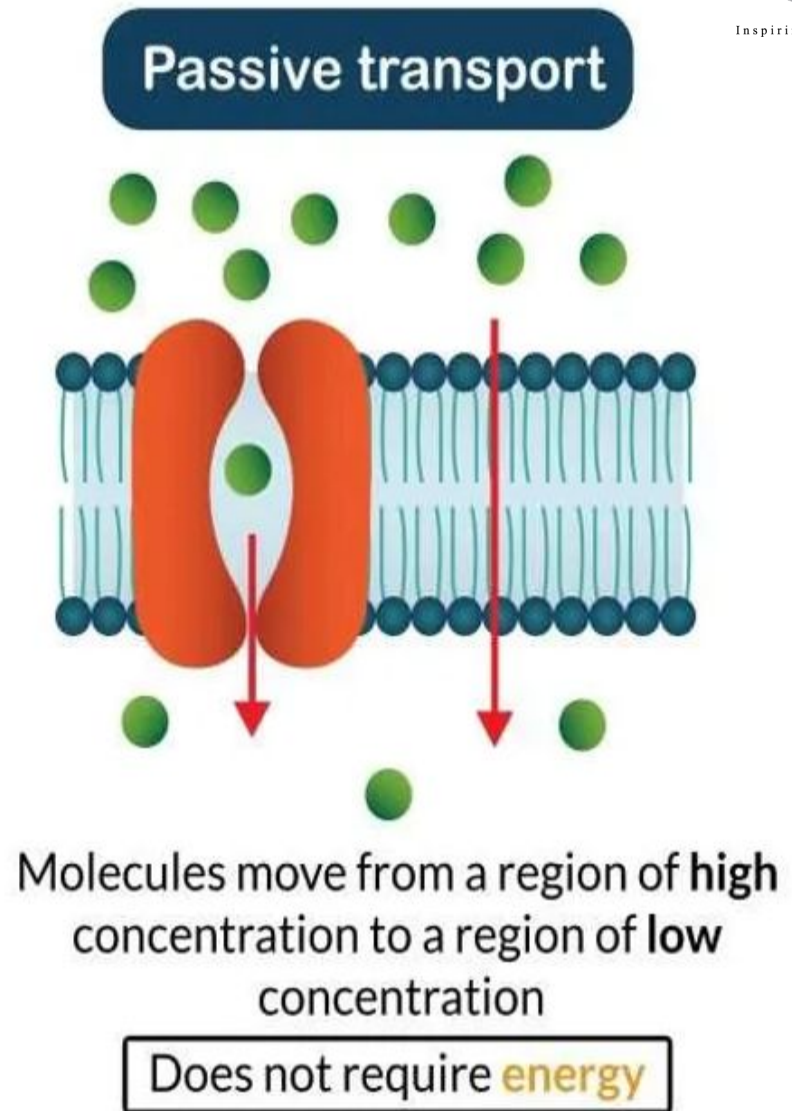


Image credit: OpenStax Biology, modified from original work by Mariana Ruiz Villareal.

Passive Transport

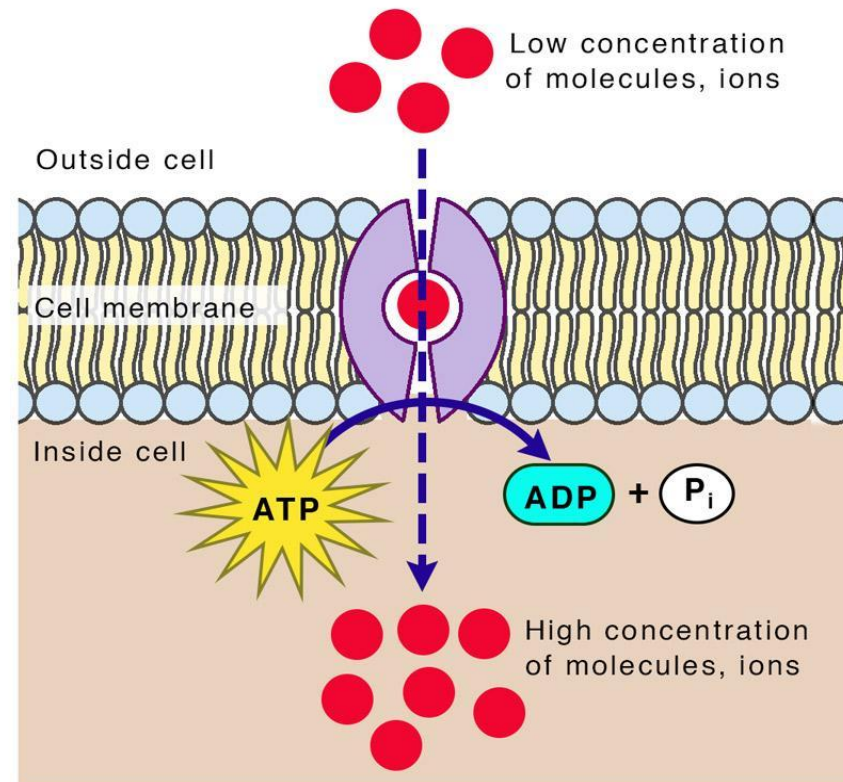
- Flow of solutes through the interior of passive transport proteins down their concentration gradients (From high to low concentration).
- Passive transport proteins allow solutes to move both ways.
- Does not require any energy input.



Active Transport

- Movement of molecules from an area of lower concentration to the area of higher concentration (against concentration gradient).
- Transport protein must be activated.
- ATP gives up phosphate to activate transport protein.
- Binding of phosphate changes transport protein shape and affinity for solute.
- The other side opens, solute is released
- Removal of phosphate
- Protein goes back to its original state

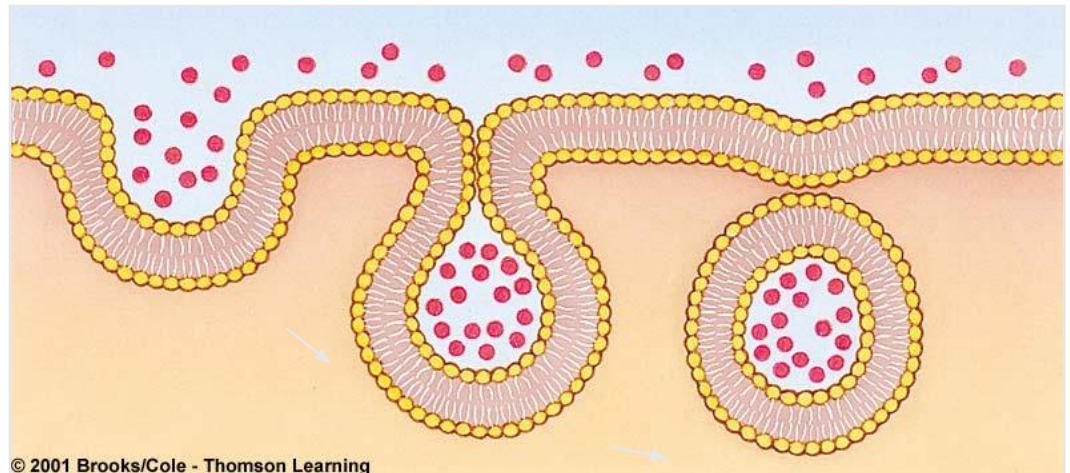
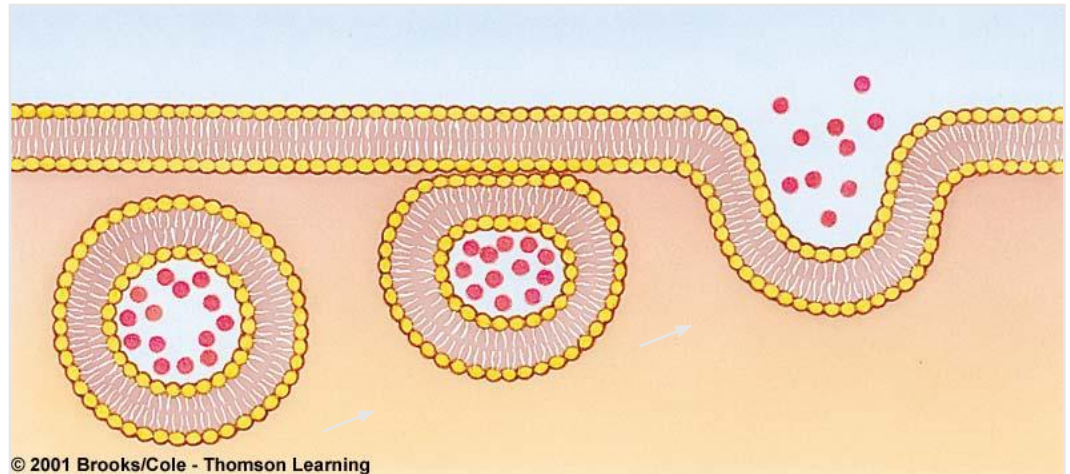
Active Transport



Requires energy

Bulk Transport

- **Exocytosis:**
Substances are taken out of the cell in a membrane bound vesicle.
- **Endocytosis:**
Substances taken into the cell in a membrane bound vesicle



How Cells Release Stored Energy

Cellular Respiration

- Pathway that breaks down an organic molecule to form ATP.
- In this process, organisms take food molecules, break them down and release the chemical energy stored in the chemical bonds of those molecules.
- The energy that is released from chemical bonds during cellular respiration is stored in molecules of ATP
- Example: Cells convert sugar into energy

Aim:

- To create ATP to power cellular reactions. Cells require fuel and an electron acceptor which drives the chemical process of turning energy into a usable form (ATP).

ATP is called the energy currency of all biochemical processes!

What type of organisms undergo cellular respiration?

Animals, plants, fungi, algae and other protists (also in prokaryotes in cytoplasm)

What type of molecules are broken down?

- Any food (organic) molecule, or nutrient, including carbohydrates, fats/lipids and proteins can be processed and broken down to produce ATP.



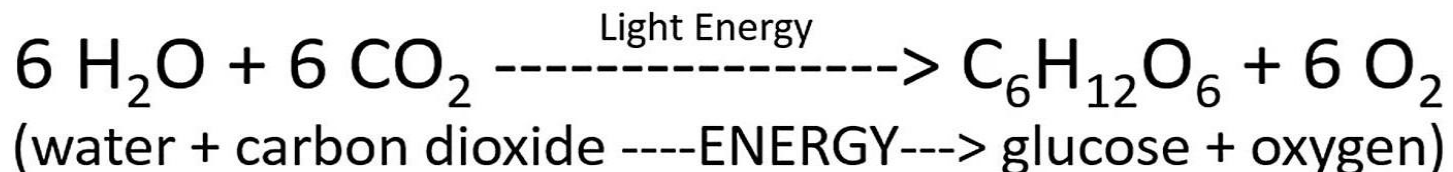
Stages of Cellular Respiration/ Aerobic Respiration

1. Glycolysis
2. Pre-Krebs Cycle (Pyruvate to Acetyl Co-A)
3. Citric Acid Cycle/TCA cycle/Krebs Cycle
4. Electron Transport Chain/ Oxidative Phosphorylation



Simple equation of cellular respiration

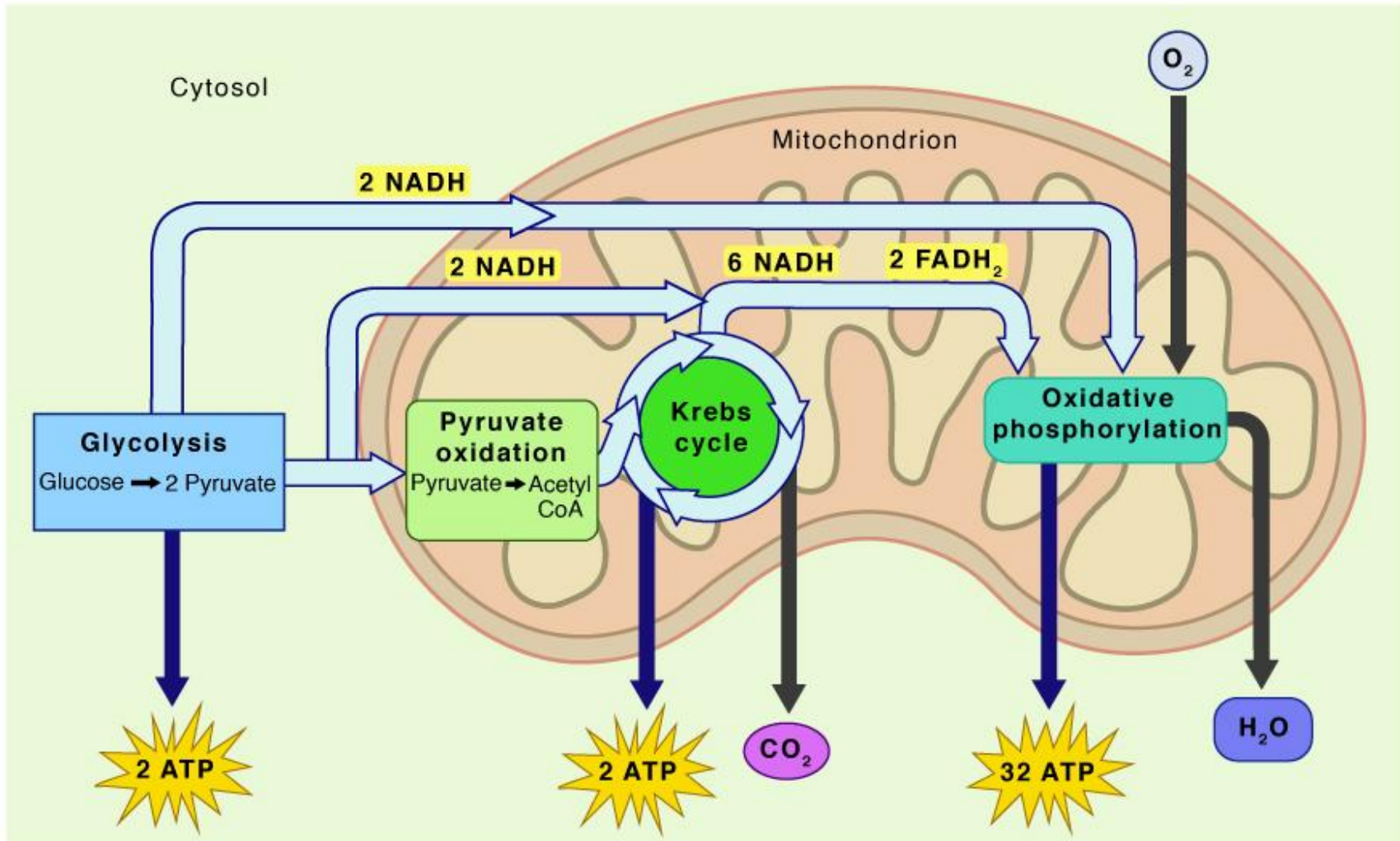
Photosynthesis



Cellular Respiration/ Aerobic Respiration

- Overall, cellular respiration is a process that is **aerobic**.
- Aerobic means that it requires the **presence of oxygen**.
- One step: **Glycolysis**, of cellular respiration do not require the presence of oxygen and is therefore an **anaerobic step**. This is the only stage that is **completed in cytoplasm**.
- **Rest of the stages** are completed in mitochondria (powerhouse: produces majority of cell's ATP).
- Energy carriers found within this process: **ATP**
- Electron carriers found within the process: **NADH, FADH₂**

Aerobic Respiration



Glycolysis

- **First stage** of aerobic respiration.
- Set of reactions that produce ATP by converting one glucose to two pyruvates.
- All reactions are mediated by enzymes.
- **Anaerobic** step in the cellular respiration pathway therefore it **DOES NOT require oxygen**
- **Location:** **Cytoplasm** (For both prokaryotes & eukaryotes)
- **Product:** 2 Pyruvate, net yield of two ATP and two NADH.
- Pyruvate moves to mitochondria

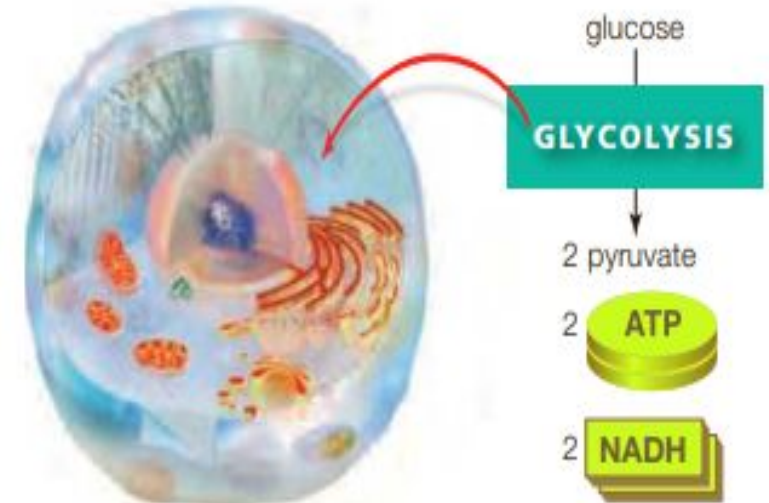


Figure Overview of glycolysis.

Glycolysis converts one molecule of glucose to two molecules of pyruvate, for a net yield of two ATP and two NADH. An animal cell is shown, but the pathway occurs in the cytoplasm of all cells, prokaryotic and eukaryotic.

Glucose		2 Pyruvate
-C-C-C-C-C-C-	→	-C-C-C- -C-C-C-
(six carbons)		(three carbons)

Pre-Krebs Cycle (Pyruvate to Acetyl-CoA)

- Before a **pyruvate** enters the Krebs's Cycle, it combines with a coenzyme called **Coenzyme A (CoA)**.
- This reaction produces **one molecule of Acetyl CoA from one pyruvate**.
- **Acetyl CoA** is a molecule produced by almost all nutrients (carbohydrate, protein, lipid) before entering the Krebs's cycle.
- **Location:** **mitochondrial matrix** (eukaryotes), **cytoplasm** (prokaryotes)
- **Product:** Acetyl CoA; CO_2 released, NADH produced.

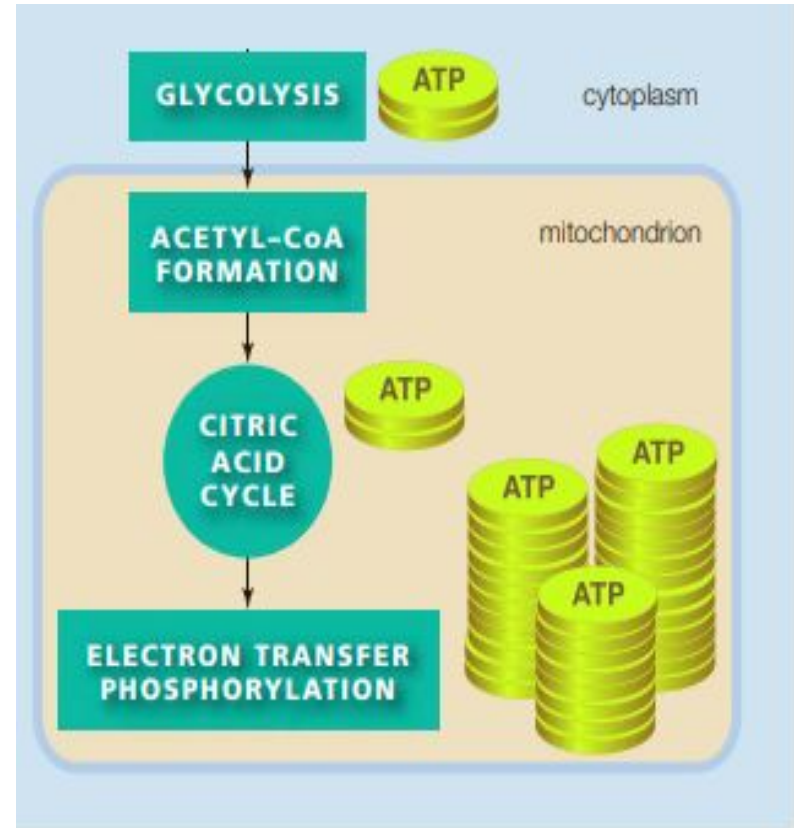


Fig: Pathway to Citric Acid Cycle

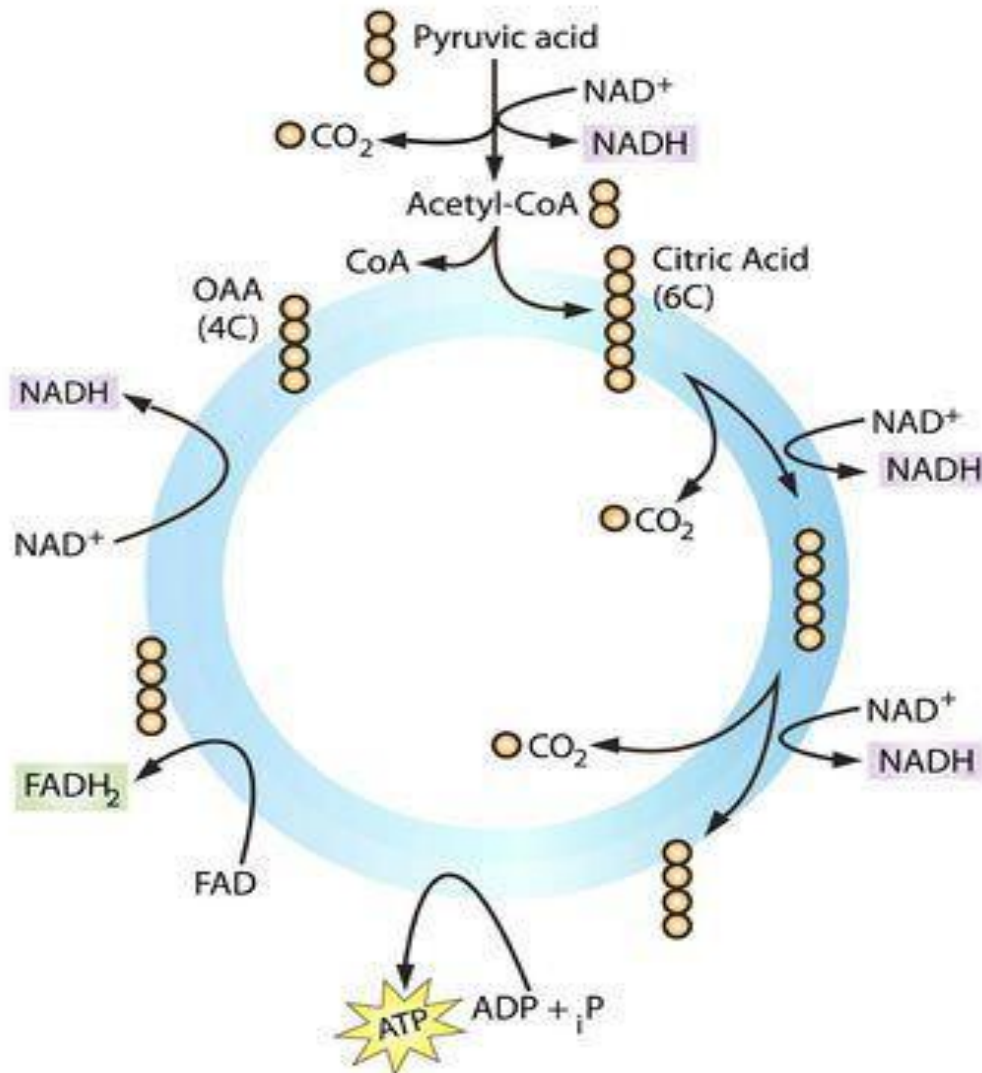
Krebs Cycle

- Named after Hans Krebs.
- Requires **oxygen (aerobic)**.
- An Acetyl CoA (formed from Pyruvate) combines with a four-carbon molecule (OAA) to make one molecule of citric acid: starting molecule of Krebs's cycle.
- Citric acid is broken down in several steps providing the energy to make NADH, FADH_2 , ATP.
- **Cyclical series** of oxidation reactions that give off CO_2 and produce one ATP per cycle.
- Turns twice per glucose molecule (produces 1 ATP per turn).
- Produces $(2 \times 1 \text{ ATP}) = 2 \text{ ATP}$ molecules from 1 glucose molecule.
- **Location:** **Mitochondrial matrix** (Eukaryotes), **Cytoplasm** (prokaryotes)
- **Product:** NADH, FADH_2 , ATP; CO_2 released.

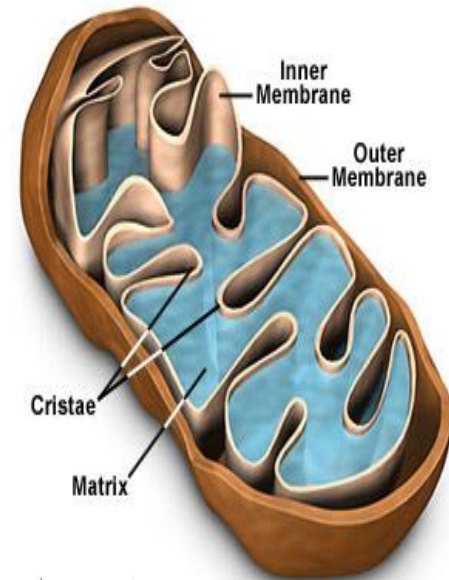


Hans Adolf Krebs (1900-1981)

Krebs Cycle (Citric Acid Cycle)



Mitochondria Structural Features



Product summary/ glucose molecule

- 2 ATP $\longrightarrow$ Usable energy
- NADH $\longrightarrow$ Goes to ETC
- FADH_2 $\longrightarrow$ Goes to ETC
- CO_2 $\longrightarrow$ By product

Electron Transport Chain

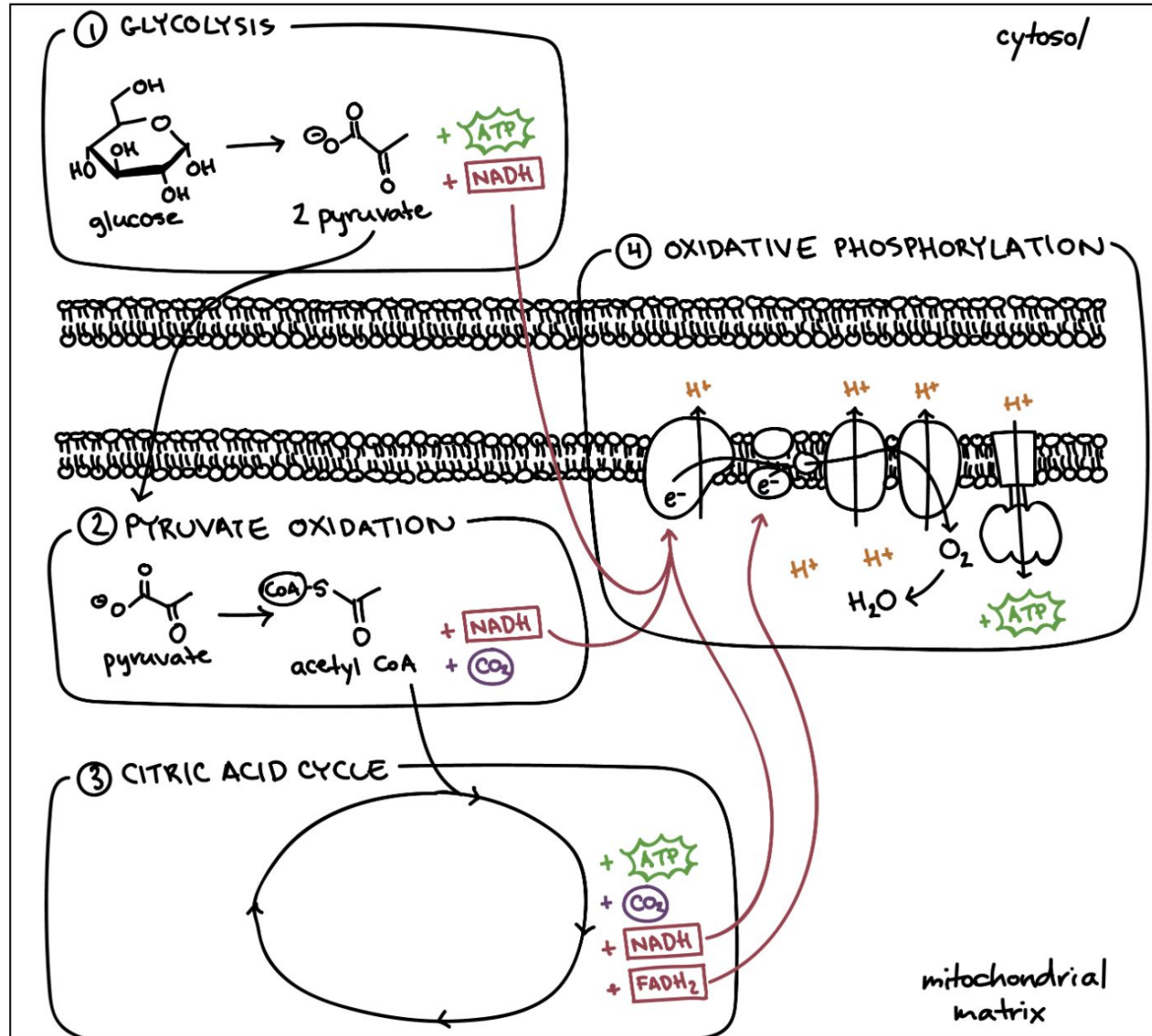
- The ETC is a series of proteins located in the inner mitochondrial membrane.
- It accepts high energy electrons from the NADH and FADH_2 provided by the previous steps.
- As the electrons move down the chain, energy is released. This energy is used to move H^+ (proton) out of matrix which creates a concentration gradient.
- These H^+ flows down the concentration gradient, back into matrix through ATP synthase to form ATP.
- **Oxygen** is used as the **final electron acceptor** at the end of the ETC.
- Oxygen receives electrons and hydrogen ions (H^+) and produces H_2O .
- Location: **Inner mitochondrial membrane (Eukaryotes)**
Cytoplasmic membrane (Prokaryotes)

Electron Transport Chain

ETC Product Summary

• **34 ATP** → Usable energy

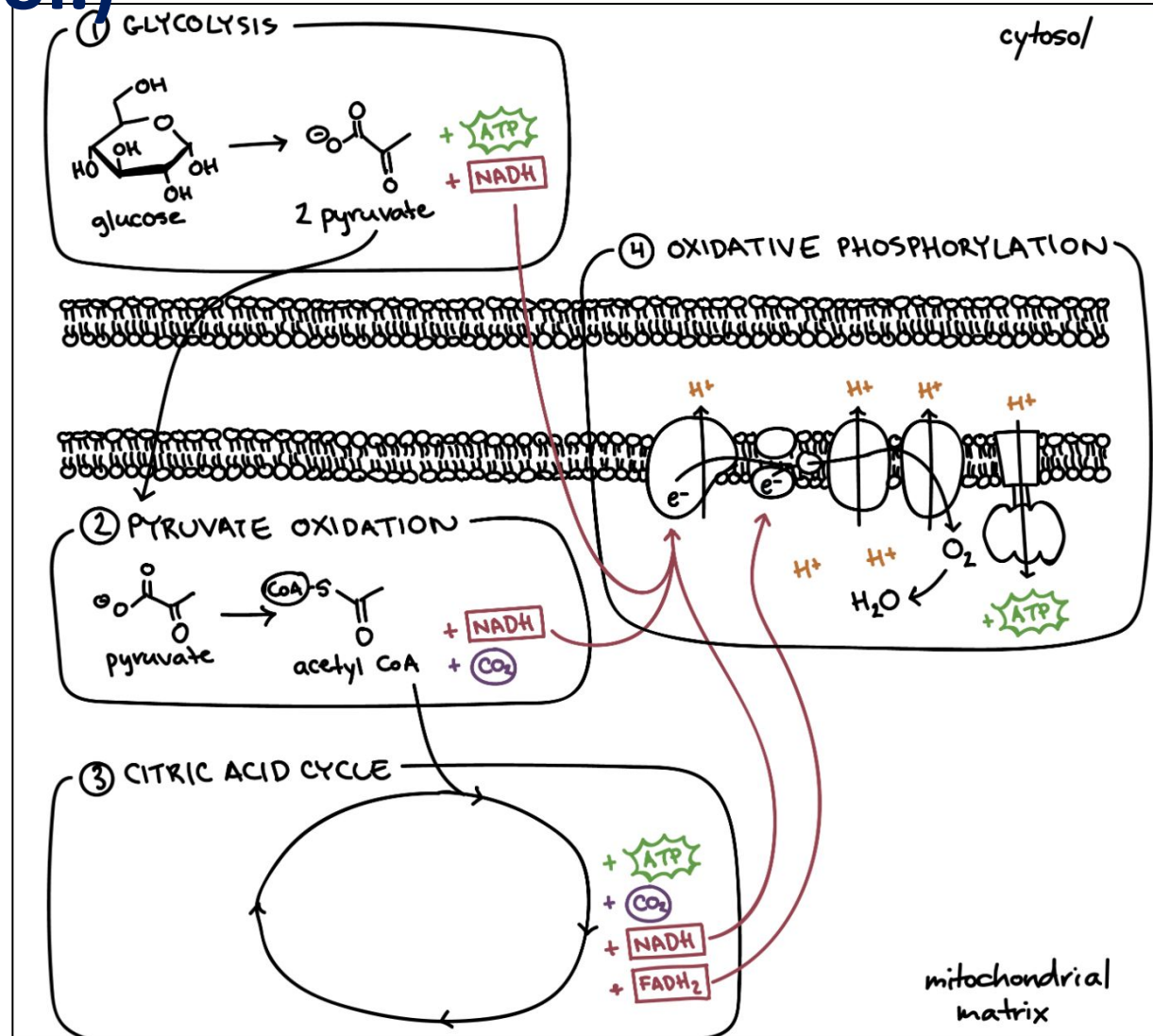
• **H₂O** → By product



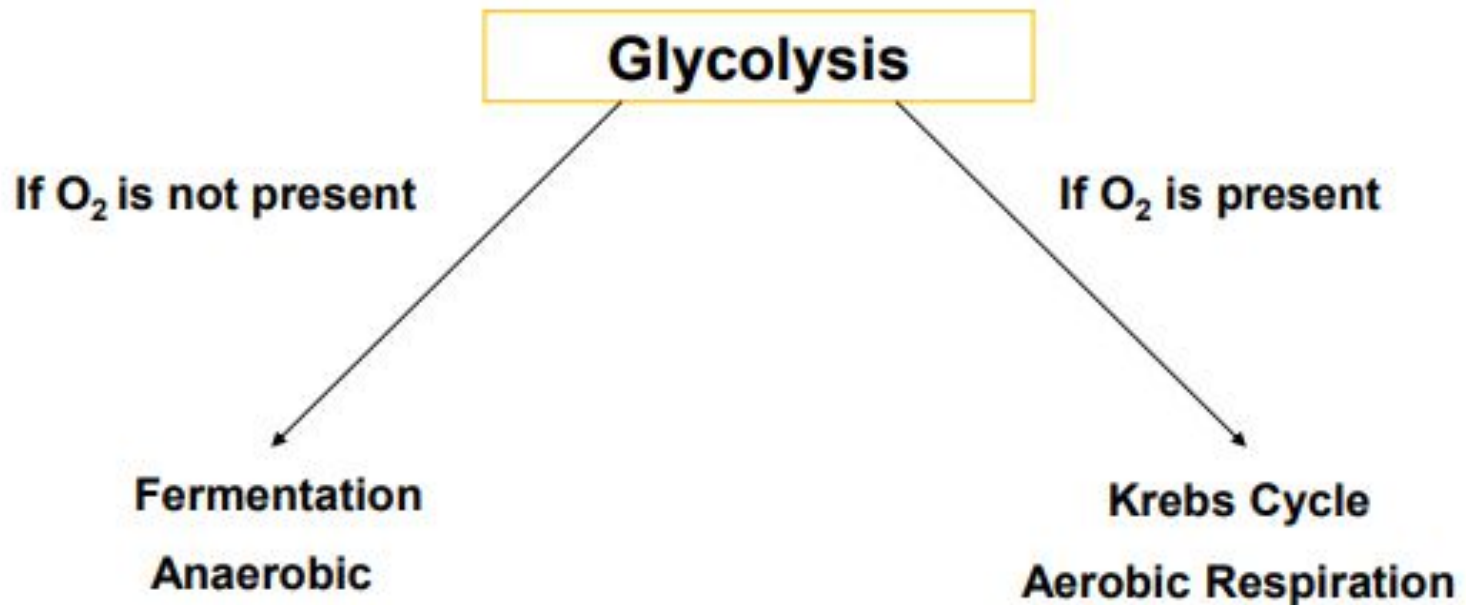
Cellular Respiration ATP Tally (for 1 glucose mol.)

1. Glycolysis = 2 ATP
2. Krebs Cycle = 2 ATP
3. ETC = approx. 32 ATP

Grand Total = 36 ATP



What happens after glycolysis, if oxygen is not present?



Fermentation

- Glucose-breakdown pathways, that make ATP without the use of oxygen or electron transfer chains.
- If **oxygen is not present**, the products of glycolysis (pyruvate and NADH) will enter an alternative process called fermentation.
- Like aerobic respiration, fermentation **begins with the glycolysis** pathway (anaerobic step).
- Uses **other molecules as a final electron acceptors** (instead of O_2)
- Location: **Cytoplasm**
- Two types of fermentation:
 - **Lactic Acid Fermentation** = Humans (muscle cells)
 - **Alcoholic Fermentation** = Yeasts (bread making)

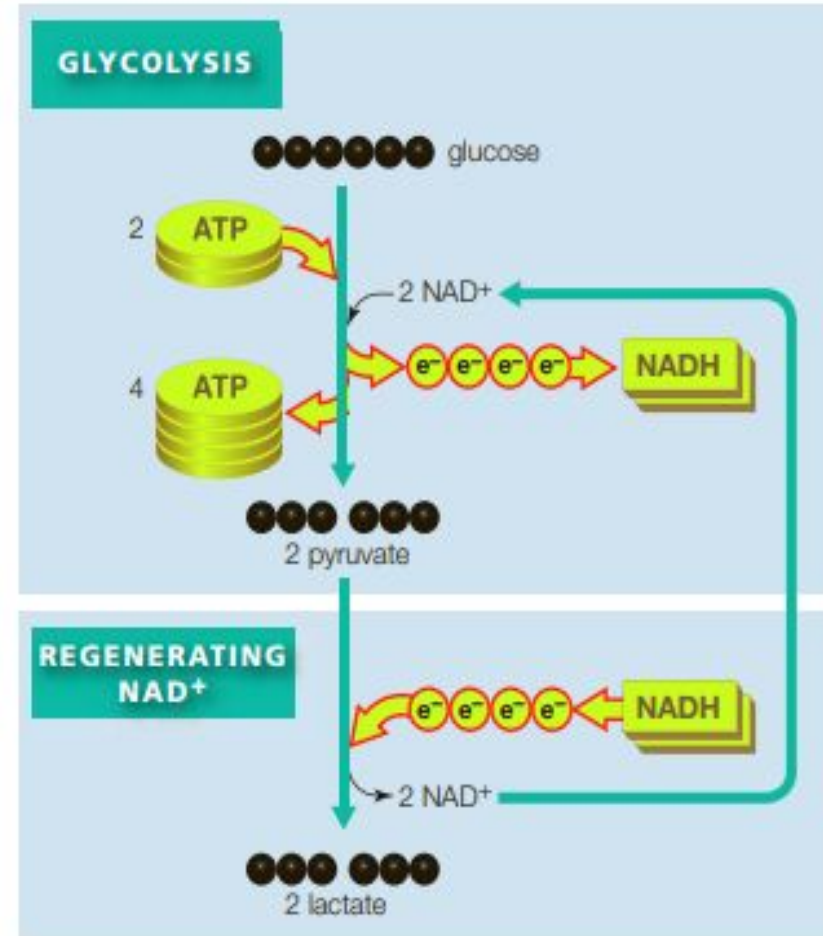
Examples of Fermentation

Lactic Acid fermentation

Breathing provides enough oxygen for body to carry out normal activities. But during high level activity, breathing can't supply enough O_2 to muscle cells.

So, the **muscle cells** switch to fermentation (anaerobic process) to generate ATP, by producing lactic acid/ lactate.

- **Electron acceptor: Pyruvate**
- **Location: Cytoplasm**
- **Product: Lactic acid, ATP (generated only in glycolysis)**



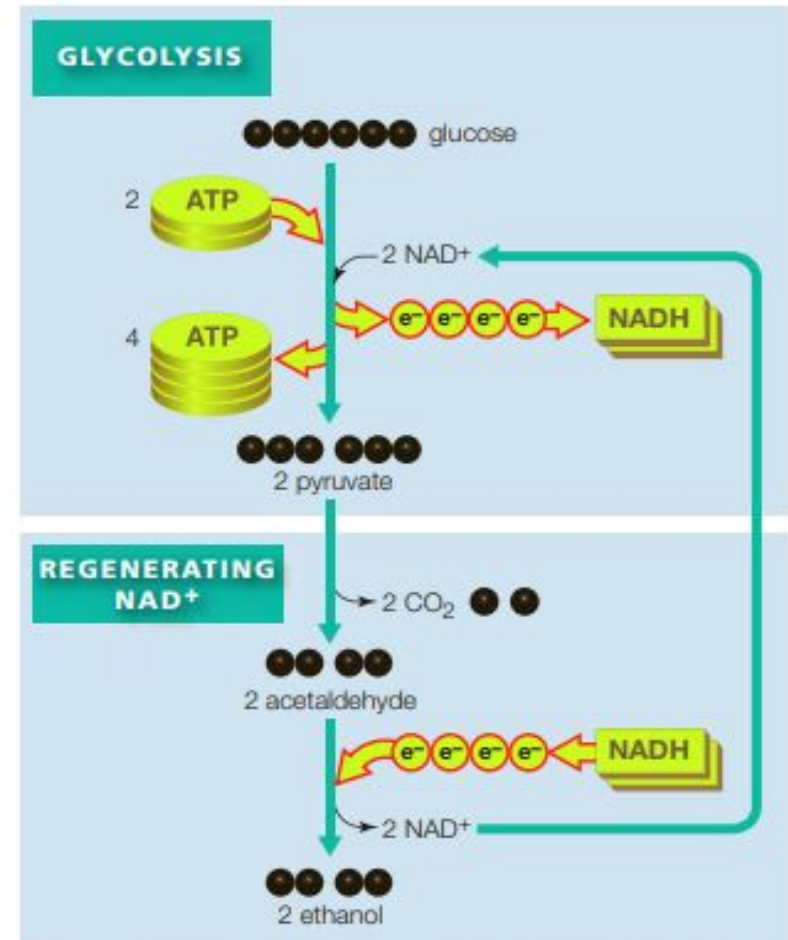
A Lactate fermentation begins with glycolysis, and the final steps regenerate NAD^+ . The net yield of these reactions is two ATP per molecule of glucose (from glycolysis).

Examples of Fermentation

Alcoholic Fermentation

Saccharomyces cerevisiae, also known as brewer's or baker's yeast, converts sugars and starches into alcohol and carbon dioxide during the fermentation process.

- Electron acceptor: Acetaldehyde
- Location: Cytoplasm
- Product: Ethanol, ATP (generated only in glycolysis)



A Alcoholic fermentation begins with glycolysis, and the final steps regenerate NAD^+ . The net yield of these reactions is two ATP per molecule of glucose (from glycolysis).

Aerobic Respiration	Fermentation
Oxygen is required to derive energy.	Oxygen is not required in this process.
Site of occurrence : Cytoplasm & Mitochondria.	Site of occurrence: Cytoplasm.
End products are CO ₂ & water.	End products are lactic acid; ethanol, CO ₂ .
Involvement of electron transport chain & Krebs's cycle.	No electron transport chain & Krebs's cycle.
Final electron acceptor: O ₂ .	Final electron acceptor: Organic molecule (e.g Pyruvate/ Acetaldehyde).
About 36 molecules of ATP are produced.	Only 2 molecules of ATP are generated (during glycolysis)
Mainly occurs in higher organisms.	Mainly occurs in microorganisms.

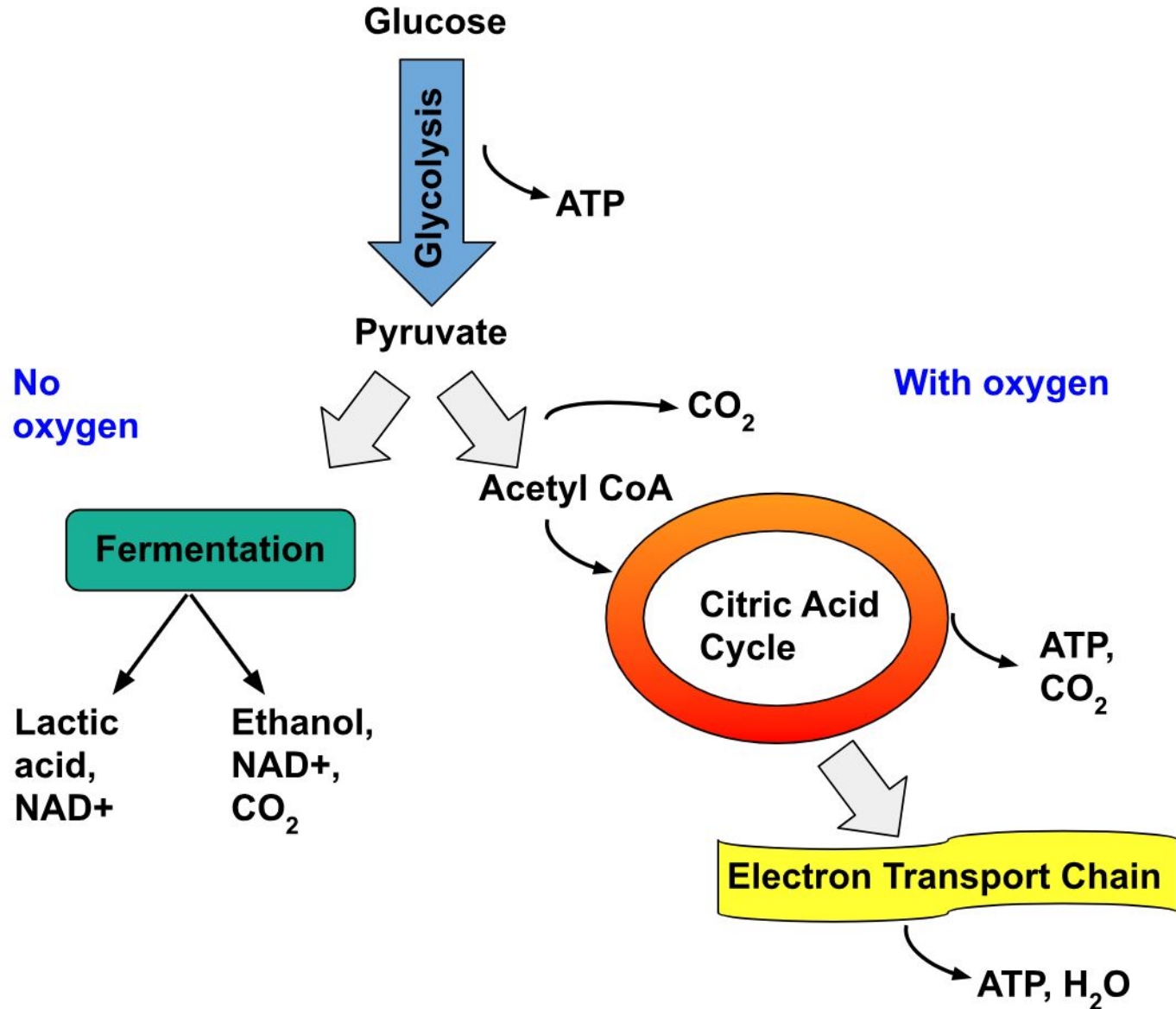
Advantages of Fermentation

- **Increases habitat:** It lets organisms live in places where there is little or no oxygen. Such places include deep water, soil, and the digestive tracts of animals such as humans, some bacteria live in the human digestive tract.
- **Speed:** It produces ATP very quickly. For example, it lets muscle cells get the energy they need for short bursts of intense activity. Aerobic respiration, on the other hand, produces ATP more slowly.

Aerobic Respiration vs Fermentation

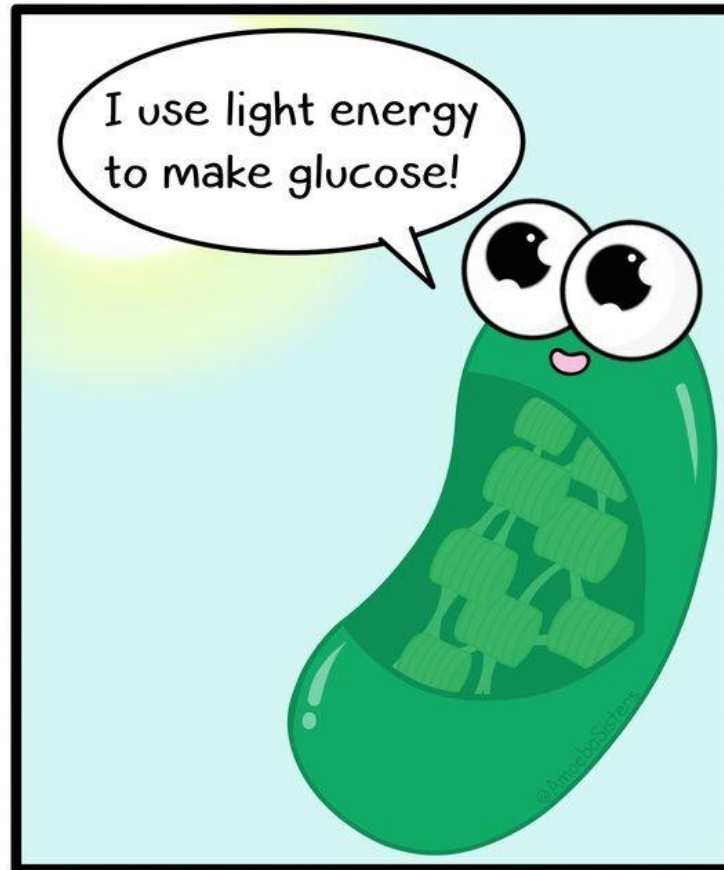
- Aerobic respiration produces much more ATP than fermentation.
- Fermentation occurs more quickly than aerobic respiration.

Overview

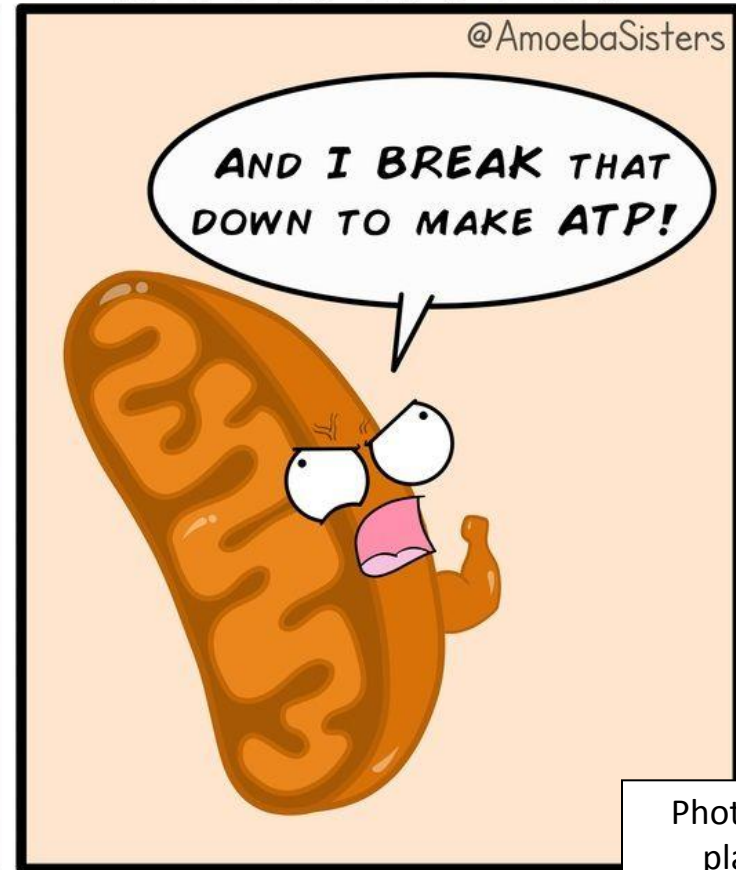


Thank You!

CHLOROPLAST VS MITOCHONDRION



- Found in (most) photosynthetic eukaryotes
- Used for photosynthesis



- Found in (most) eukaryotes
- Used for cellular respiration

Photosynthetic plants too!